

A National Airspace System Reference Architecture as a System of Systems

Tracing Operational Intent from Enterprise Objectives to Aircraft
Trajectory

ISD 503 Capstone Project Report – Interim Report #1

Submitted to:

Mark Petrotta - Principal Engineer of Digital Engineering, MBSE, and Augmented Intelligence -
Aerospace and Defense Systems

IOE Department of Integrated Systems Engineering & Design

Prepared by:

Owen O’Keefe

University of Michigan, ISD

September 19, 2026

Plan ref: to-do-list.md S1 — Confirm advisor/reader name and title for "Submitted to."

Executive Summary

Plan ref: to-do-list.md S14, S16 — Write this section LAST. Distill: (1) the improvement opportunity/problem statement, (2) the MBSE approach used, (3) major architecture and optimization-demonstration findings stated quantitatively, (4) the operational impact/benefit of the reference architecture (see Section 5.0 Impact/Financial Benefits), and (5) recommendations. Limit to 1-2 pages; no citations, undefined terms, or content not found elsewhere in the report.

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1 Introduction and Current State

Plan ref: to-do-list.md S1 — Briefly state the subject, purpose, and scope of the report: the NAS modeled as a System of Systems, centered on how operational intent propagates from enterprise objectives to aircraft trajectory/behavior.

My project aims to build a reference MBSE architecture for how airspace intent is managed across the NAS — spanning the organizations that define and enforce rules, the domains managing airspace resources, the systems supporting flight operations, and the onboard systems that turn intent into executable trajectory and guidance behavior.

In his 2005 paper, DeLaurentis states the following DeLaurentis [8]:

“Problems of significant complexity are facing decision-makers within government and industry, and the moniker of ‘system-of-systems’ (heretofore shortened to SoS) is increasingly being applied to problems of this type. Effective system analysis for decision-support quickly becomes unmanageable in this context, with multiple, heterogeneous, distributed systems involved and couched in networks at multiple levels. The situation is exacerbated by the fact that most organizations (and their processes) remain in the ‘stovepipe’ mode – their people and tools are configured to study within narrow bins with little or no analysis across those bins. Current frames of reference, thought processes, analysis, and design methods are not complete for these SoS problems ... A holistic framework is needed that enables decision makers to discern whether related infrastructure, policy, and/or technology considerations together are good, bad (or in-different) over time.”

While this reference is from 2005, it is still relevant today. The NAS is a complex system of systems, and the decisions made by the various stakeholders have significant consequences for the safety, efficiency, and sustainability of the air transportation system.

Airline companies invest significant time and money in constructing operational plans that maximize revenue; however, on the day of operation, unpredictable events – bad weather, aircraft malfunctions, crew absenteeism – can substantially alter that plan and the revenue objectives behind it Castro [5]. To manage this, airlines maintain an Airline Operations Control Center (AOCC), a department of experts specialized in disruption management across aircraft, crew, and passenger recovery. The AOCC monitors operations and makes real-time decisions – re-routing flights, rescheduling crews, rebooking passengers – intended to minimize the impact of a disruption on the airline’s schedule and revenue objectives.

The literature does not converge on a single, standardized definition of the “cost” the AOCC is minimizing, and the values it does quantify do not always hold up under scrutiny. Delay cost is commonly modeled in the optimization literature as a linear function of delay minutes, but Hu et al. [22] notes that this assumption is inconsistent with airline practice, where delay cost is in fact a nonlinear function of delay time – and that, faced with this complexity, airlines tend to prioritize the speed of a recovery decision over the mathematical optimality of the resulting solution. This preference for tractable, fast decisions over globally optimal ones shows up structurally as well as behaviorally: a review of the Aircraft Recovery Problem finds that studies – and, by extension, the AOCC processes they model – rarely minimize delay, cancellations, and aircraft swaps simultaneously, instead treating aircraft, crew, and passenger recovery as separable sub-problems solved sequentially and handed off in turn Santana et al. [43]. Each stage can produce a solution that is locally optimal yet globally suboptimal, or even infeasible, once the next stage’s constraints are applied – a concrete, literature-documented instance of the myopic-optimization pattern this capstone investigates.

That pattern’s costs and benefits are visible even in work that improves on the status quo. Kontodimou et al. [28]’s predictive turnaround-buffer allocation model achieves a 37.1% reduction in propagated delay relative to a zero-buffer baseline, but it reaches that result by holding aircraft routing, fleet assignment, crew duty, maintenance, and passenger connectivity fixed as upstream constraints, deliberately excluding them from the optimization it performs. The gain is real within that bounded scope, but the framing illustrates the exact risk this capstone is concerned with: a locally-optimized decision layer can report a strong local benefit while externalizing tradeoffs onto adjacent decision layers that go unmeasured by that layer’s own cost model. The same pattern appears on the human side of the AOCC: Turhan, Gzara, and Elhedhli [49] documents that when dispatcher workload exceeds capacity, dispatchers “may resort to conservative approaches, which can have financial implications for airlines” – a workload-driven cost that rarely appears in the delay, cancellation, and swap cost models discussed above. Hassan, Santos, and Vink [16] frames this more broadly as a persistent gap between the disruption-management optimization literature and the reality of AOCC practice.

In this paper, I explore the value tradeoffs that aviation stakeholders experience when disruptions occur and recovery solutions are employed with various levels of myopia. When a dispatcher or AOCC team member makes a decision, they are often focused on the immediate impact of that decision – e.g., minimizing delay for a specific flight – without visibility into the broader implications for the airline’s overall operations or the NAS as a whole. Cost models used today typically capture the factors that are easiest to quantify, such as fuel consumption and crew costs, while factors that are harder to quantify – maintenance cost, passenger satisfaction, dispatcher/controller workload, environmental impact – are more often overlooked. This can lead to suboptimal outcomes: increased congestion in certain airspace sectors, missed opportunities for more efficient routing, or decreased overall profits driven by customer dissatisfaction. By modeling the NAS as a system of systems and analyzing the propagation of operational intent, I aim to identify strategies for improving decision-making and reducing the negative impacts of disruptions on the air transportation system.

1.1 Overview of the Project

Plan ref: to-do-list.md S1 — Describe the industry/environment (commercial aviation and the NAS), the social context, and the key stakeholders (preview – full analysis in Section 1.3). State the project’s research questions from Project_To-Do List.md S1 (“Finalize project research questions”) and explain why they matter. Draw on README.md’s framing of the trajectory-intent lifecycle (Strategic objective -> mission plan -> flight plan -> ATC constraints -> trajectory negotiation -> FMS intent -> guidance commands -> aircraft motion) as the throughline.

The NAS is a complex system in which operational decisions are made by independent actors, each of which may or may not be accounting for the impacts on other system stakeholders, which includes regulatory agencies, air traffic service providers, airport operators, airline operations centers, MRO suppliers, flight crews, passengers, and aircraft systems. While substantial architectural work has been performed within individual domains, the relationships and information flows between these domains are often incompletely represented.

Existing models frequently focus myopically on one actor or problem, exhibiting one or more of the following limitations: - Emphasis on enterprise-level capabilities without sufficient representation of aircraft behavior and onboard decision-making, - Emphasis on aircraft systems without adequate representation of the broader operational context within which those systems operate, - limited traceability between strategic objectives, operational constraints, human decisions, and the resulting aircraft trajectories, - difficulty evaluating the impacts of new operational concepts,

automation technologies, and optimization services across organizational boundaries.

As a result, there is no broadly applicable architecture that fully captures how strategic objectives become executable aircraft behavior. This lack of end-to-end traceability limits the ability of engineers, operators, and decision makers to understand the consequences of changes within the NAS and complicates the integration of future decision-support technologies.

This project seeks to address this gap by developing a System-of-Systems architecture that explicitly models the propagation of intent throughout the National Airspace System, from enterprise objectives and airspace management decisions to flight deck decision-making and aircraft trajectory execution.

1.2 Review of Current State Design / Literature Review

Plan ref: to-do-list.md S2–S4 — Synthesize (do not merely list) findings from the three literature-review tracks: airline planning and operations (schedule planning, fleet assignment, crew scheduling, maintenance routing), day-of- operations and turnaround (A-CDM, weight and balance), and OCC/dispatch and flight execution (disruption management, dispatcher responsibilities, nominal ATC/IFR flow). Pull from per-source annotations captured via the literature-review workflow into `evidence/source-register.md` rather than re-deriving them here. Identify the research gap this capstone addresses (see also `Project_To-Do List.md S16` "Research gap").

From my perspective, the current state of the project is off-track, but recoverable. The plan calls for 4 phases of work, with phase 1 completing 9/10 with RACII matrices for the various objectives within the airspace. More time has been spent on the literature review than originally expected, as the scope broadened to encompass consideration of which ontological model to use. Moreover, a lack of discipline on extracting specific actionable information from the review has weakened definitive claims about specific objectives pursued by stakeholders, which in turn feed the considerations of the RACII matrix. While much learning has occurred from the literature review, the impact on schedule continues. Presently, 53 papers have been extracted, with more than half manually reviewed, with many annotated. The literature review is ongoing, but the current content supports the research claims and enforces the value of the project deliverables.

Time spent developing the Concepts of Operations has also impacted other work. However, the project lacks a specific focus at the moment for what tradeoffs will be considered. While the project does not intend to be an exploration of optimization, the concept focuses on sub-optimality driven by myopic decisions. While the concepts of operations help inform a more comprehensive understanding of the NAS, the systems involved, and the interfaces amongst them, it has not yet provided for unique insights into specific scenarios that might be worth modelling.

At this stage, the progress is at risk of analysis paralysis, which is being weighed against the risk of a lack of a well-formed vision for execution. At this point, I wish to heir on the side of action, transitioning the project to Phase 2 and allowing specific gaps or needs in the model development guide further review of the collected sources. That said, I still plan to continue the review ad-hoc. Also worth noting is that model development will inform possibilities w.r.t. what can be modelled (e.g. multiple instances of an aircraft system performing similar, but different actions simultaneously while each affecting the same controller's workload). Additionally, gaining experience with `sysmlv2` will inform whether its learning curve makes it a viable tool, or if Cameo is more appropriate.

This pivot from Phase 1 exploration to Phase 2 modelling will advance the project deliverables while engraining a deeper understanding of the information gaps. When the modelling gets to the point of MOEs/MOPs, the objectives of stakeholders will be revisited and a more targetted literature consultation can occur.

Per the original plan, state machines for the various NAS systems were intended to be completed

by 9/15. This deliverable has not been met. The realities of a full-time professional job whose needs often extend beyond the normal 40-hour work week has forced much of the project work to occur on weekends and my time management needs to adjust to accommodate. This weekend will be a decision gate that informs that process and possibly lead to an adjustment of the schedule of deliverables. (Section 1.5).

1.3 Stakeholders

Plan ref: to-do-list.md S7 — Identify key project stakeholders (individuals/organizations affected by or perceiving themselves affected by NAS operational decisions). Draw on the PESTLE stakeholder discovery, stakeholder register, and RACCI matrix from Project-To-Do List.md S7. Confirm the stakeholder set is diverse (airline, passenger, ATC/ANSP, airport, flight crew, environmental/ societal, military – see S8’s per-stakeholder objective categories) and that stakeholders with little formal power (e.g., passengers, communities affected by noise/emissions) are represented.

I created the extttcapstone extunderscore stakeholder extunderscore assessment.xlsx file, filling it with my initial assessment of each candidate stakeholder’s objectives, and then iteratively refined it as I read more literature and gained a better understanding of the NAS. The assessment includes the following information for each stakeholder:

- stated interest and whether it acts as a hard constraint, optimization objective, cost/penalty, MOP, or MOE
- candidate explorations of MOP/MOEs
- determination of whether the stakeholder has a direct or indirect impact on trajectory decisions, if any.
- An exploration of whether the stakeholder should be a standalone stakeholder or consolidated with another stakeholder, and the rationale for that decision.

This has lead me to the following list of stakeholders that I believe are relevant to the project:

- Governments / Standards Organizations
- Operators (Airlines, Charter Operators, Cargo Operators)
- Air Traffic Control (ATC)
- Aircraft and Engine Manufacturers
- Airport Authority
- Flight Crew
- Passengers
- Environmental/Societal Organizations

Furthermore, an exploration of the interests of each stakeholder has lead me to the following list of objectives that I believe are relevant to the project:

- Safety / Security
- Efficiency

- Throughput / Capacity
- Reliability / Resilience
- Environmental Impact
- Noise Impact
- Passenger Satisfaction
- Schedule Reliability
- Cost / Profitability
- Maintenance / Lifecycle Cost
- Regulatory Compliance
- Workload / Human Factors

1.4 Scope

Plan ref: to-do-list.md S1 — State what is explicitly included and excluded from the System of Interest (SOI), the level(s) of abstraction modeled, and the criteria used to draw system and stakeholder/actor boundaries. This section should resolve the open boundary questions currently tracked in `knowledge/questions/open-questions.md S1` – if any remain open when this section is drafted, that is a signal the SOI work in `Project-To-Do List.md S1` is not yet closed.

See Section 1.1 for a discussion of the scope of the project. Also note ambiguities highlighted in the review of the current state design/literature review (Section 1.2) that will be addressed in future phases of the project.

1.5 Project Deliverables

Plan ref: to-do-list.md S10, S14, S16 — State the goals and deliverables: the SysML/Cameo reference architecture, the traceability chain from stakeholder needs to aircraft-level behavior, and the optimization-demonstration capability (Section 3). Include the quantifiable measures used to assess success (e.g., traceability coverage, stakeholder coverage, demonstration of a local-vs-SoS optimization tradeoff).

The following deliverables were established for the project upon its inception:

See the review of the current state design/literature review (Section 1.2) for a discussion of the current status of these deliverables and the risks to their timely completion.

1.6 Project Timing

Plan ref: to-do-list.md S1 (new) — Revisit the timeline from the 503 Proposal submittal (see `prework/ISD 503 Submittal.pdf`) and list the sequence of tasks – keyed to `Project-To-Do List.md`'s 16 sections – with target completion dates. A "Project Timing" task did not previously exist in the plan; it has been added to `Project-To-Do List.md S1` this session so this subsection has something concrete to draw from.

Table 1: Project milestones and verification methods

Project Milestone	Deliverable	Lead/Team Member	Expected Completion Date	Target Metric/Verification Method
Literature Review	Annotated bibliography	Owen	9/7/2026	10–20 papers categorized
PESTLE Analysis & Systems-of-Interest Definition	Context diagram, text-based documentation	Owen	9/10/2026	Review
Systems-of-Interest Definition	Text-based documentation	Owen	9/15/2026	Clear boundaries and exclusions
Operational Scenarios	ConOps document	Owen	9/22/2026	1–3 representative scenarios
Draft Architecture	BDDs and package structure	Owen	10/1/2026	All domains represented
Interface Definition	Interface Control Document	Owen	10/12/2026	All interfaces identified
Internal Block Diagrams	IBDs	Owen	10/17/2026	Critical interfaces connected
Requirements Definition	Requirements model	Owen	10/20/2026	Trace to stakeholders
Measures of Effectiveness	MOE/MOP hierarchy	Owen	10/25/2026	Quantifiable metrics defined
Requirement Traceability	Requirements matrix	Owen	10/29/2026	Complete allocation
Activity Diagrams	Operational activities	Owen	11/3/2026	Scenario walkthrough
Sequence Diagrams	Information exchanges	Owen	11/8/2026	Message completeness
State Machines	State behavior of key entities	Owen	11/13/2026	Executable state logic
Authority & Responsibility Models	RACII (Responsible, Accountable, Consulted, Informed)	Owen	11/17/2026	All decisions assigned
Optimization Prototype	Working code	Owen	11/21/2026	Unit test cases
Integration into MBSE Architecture	Traceability links	Owen	11/24/2026	End-to-end scenario
Final Baseline & Final Paper	Versioned model & thesis/paper	Owen	12/16/2026	Draft
Paper Submittal	Thesis/paper	Owen	12/20/2026	Official submittal

2 Assumptions and Methodology

Plan ref: to-do-list.md S1 — State the working assumptions carried through the project – see README.md "Working Assumptions" and `projects/nas-sos-capstone/index.md` "Working assumptions (guardrails)" (e.g., emphasis on structure/behavior/interfaces/traceability rather than optimization math; decision-support/optimization treated as a capability inside the architecture, not the whole subject).

Plan ref: to-do-list.md S6 — Describe the methodology for exploring and comparing alternative system decompositions (organization/authority-based, lifecycle/process-based, physical-system, information-flow, decision-authority, intent-to-trajectory) and the criteria used to evaluate them (what each exposes/obscures, where boundaries are ambiguous, which best supports optimization-impact analysis).

Plan ref: to-do-list.md S10 — Describe the MBSE methodology used to build the architecture in Cameo: package organization, diagram types produced (BDD, IBD, activity, sequence, requirements), and the traceability approach (stakeholder needs -> objectives -> requirements -> systems -> activities -> information exchanges -> aircraft behavior). Note per `workflows/update-architecture.md` that Cameo is the source of truth and the XML export is a generated artifact.

Plan ref: to-do-list.md S11 — Describe the optimization-study methodology: scenario selection criteria, system-state and decision-variable definitions, constraint definitions, the multi-objective formulation $J = w_1 J_1 + w_2 J_2 + \dots + w_n J_n$, normalization approach for objectives with different units/scales, and the stakeholder weighting scheme. This section should be detailed enough that the scenario and formulation could be reproduced.

3 Results and Discussion

Plan ref: to-do-list.md S5 — Present the nominal-flight ConOps (strategic/commercial planning through day-of-operations, turnaround, flight execution, and postflight/continuation) as the operational backbone the architecture models. Summarize; put the full activity breakdown in Appendix A.1.

Plan ref: to-do-list.md S8 — Present the stakeholder objective/cost/value ontology: what "optimal" means per stakeholder (airline, passenger, ATC/ANSP, airport, flight crew, environmental/societal, military), and the classification of each objective as hard constraint, optimization objective, cost/penalty, MOE, or MOP. Highlight conflicting vs. aligned objectives and objectives with externalized costs.

Plan ref: to-do-list.md S9 — Present the myopic-optimization analysis: the definition of "local optimum" in the SoS context, candidate SoS measures of value, and the documented conflicts (decision maker, decision variable, local objective/constraints, affected stakeholders, externalized costs, SoS consequence). See the flagged scope note below.

Scope note (flag, not a TODO to resolve silently): Project_To-Do List.md S9 catalogs eight candidate local-vs-SoS conflicts, each with eight documented attributes (64 cells). A 15–40 page report body cannot carry all eight in full; plan to present 2–4 illustrative conflicts in the main body (favoring whichever one becomes the S11–S14 demonstration scenario) and move the remaining catalog entries to Appendix A.4.

Plan ref: to-do-list.md S10 — Present the SysML architecture: context diagram, package structure, stakeholder model, BDDs, IBDs, information-object model, interface/item-flow model, nominal-flight activity diagram, responsibility swimlanes, and requirements model with traceability. Embed the most illustrative diagrams here; move the full diagram set to Appendix A.3.

Plan ref: to-do-list.md S12, S13 — Present the optimization/simulation results: the baseline scenario, single-stakeholder vs. balanced/SoS optimization outcomes, and how simulation entities/decisions/objectives/constraints trace back to SysML blocks, properties, goals, and requirements. Show how the architecture exposes impacts a standalone optimizer would omit (S13's key point).

Scope note: Project_To-Do List.md S12 as written (Pareto fronts, full objective-weight sweeps, sensitivity analysis, tipping-point identification across the whole experiment matrix) reads as a full optimization research study layered on top of the architecture work. Per this project's own working assumption ("not an optimization paper"; optimization is a capability inside the architecture, not

the whole subject) and the 15–40 page report constraint, consider scoping S12’s demonstration to one representative scenario with a single weight sweep and one Pareto-style comparison, rather than the full matrix – see the note added to `knowledge/questions/open-questions.md` under ”Optimization study scope (S11).”

4 Recommendations

Plan ref: to-do-list.md S14, S15 (S16 bullet added this session) — Restate the main findings from Section 3 and provide recommendations for moving forward. Include the alternatives considered (e.g., alternative decompositions from S6, alternative stakeholder-weighting schemes from S11) and the criteria used to select the recommended approach. Suggest additional areas of study (candidate: architecture extensions, additional off-nominal scenarios from S15, unexplored myopic-optimization conflicts from S9’s catalog).

NOTE: Project_To-Do List.md did not previously have a distinct ”Recommendations” deliverable separate from ”Conclusions and future work” – added to S16 this session so this section has an explicit home in the plan.

5 Impact/Financial Benefits

Plan ref: to-do-list.md S9, S12 (S16 bullet added this session) — Quantify the impact of the reference architecture and its optimization-demonstration capability. Translate the S9 conflict analysis and S12 demonstration results into concrete operational terms wherever possible (e.g., delay-minutes, fuel burn, CO2, gate/runway utilization) and into US-dollar terms where a credible conversion exists (e.g., published cost-per-delay-minute figures from the literature review, Section 1.2). Also discuss the ethical context and any unintended consequences of the recommendations (equity-minded engineering: whose costs get externalized, and onto whom – this connects directly to S8’s ”objectives whose costs are externalized onto another stakeholder” and S9’s conflict catalog).

Scope flag: the 503 template expects a quantified financial-benefit statement (the template’s sample statements are dollar-savings claims from industry DMAIC-style projects). This capstone is an architecture/SoS project, not a cost-reduction project with a ”before” baseline in dollars, so a literal dollar-savings figure may not be available or meaningful. Project_To-Do List.md did not previously have a task that produces a quantified/monetized impact figure – an explicit ”Impact / quantified benefits” bullet was added to S16 this session so at least one deliverable targets this section directly. Recommend framing impact primarily in operational/systems terms (delay, fuel, emissions, predictability, tradeoff-surface shape) with a dollar estimate offered as an illustrative, clearly-caveated secondary figure rather than the headline metric.

6 Summary/Conclusions

Plan ref: to-do-list.md S15, S16 — Briefly restate the problem, approach, and findings/ recommendations, including potential benefits in relative terms if absolute figures are sensitive or incomplete (e.g., ”reduces network-level delay propagation by an illustrative X% under the demonstrated weighting” rather than an unsupported absolute claim). Cover: known architecture omissions, limitations of the selected decomposition (S6), limitations of the optimization demonstration (S12 scope note, Section 3), and opportunities for future expansion (e.g., additional off-nominal scenarios, unexplored S9 conflicts, deeper crew/maintenance-routing integration).

8.0 References

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A Appendices

Plan ref: to-do-list.md S5, S7, S10, S11 (general) — Use appendices for material too extensive for the main body or of interest to only some readers. Label each lettered (Appendix A, B, ...), and ensure every appendix is referenced from somewhere in the main body.

A.1 Full Nominal-Flight ConOps

Plan ref: to-do-list.md S5 — Full strategic-planning through postflight/continuation activity breakdown (see main-body summary, Section 3).

A.2 Full Stakeholder Register and RACCI Matrix

Plan ref: to-do-list.md S7 — Complete stakeholder register (needs, goals, responsibilities, authorities, resources, constraints, information needs, measures of value) and the full RACCI matrix around operational decisions.

A.3 Full SysML Diagram Set

Plan ref: to-do-list.md S10 — Diagrams not embedded in Section 3: full BDD/IBD set, requirements model, and complete traceability matrices (needs-¿objectives-¿requirements-¿systems-¿activities-¿information exchanges-¿aircraft behavior).

A.4 Full Myopic-Optimization Conflict Catalog

Plan ref: to-do-list.md S9 — All candidate conflicts not carried into the main-body discussion or the S11–S14 demonstration scenario, each with: decision maker, decision variable, local objective/constraints, affected stakeholders, externalized costs, SoS consequence, and information needed to recognize the consequence.

A.5 Optimization Experiment Detail

Plan ref: to-do-list.md S11, S12 — Full experiment matrix, constituent-model validation notes, and any sensitivity-analysis detail not included in Section 3.

A.6 Source Register

Plan ref: to-do-list.md S1–S4 — Reference or reproduce the evidence/source register (evidence/source-register.md): source, publication type, NAS lifecycle phase, systems/stakeholders addressed, architectural and optimization evidence extracted, and confidence/limitations.